Chapter 15

Stabilized Methods

In our previous discussion of second-order differential operators, we have seen the significance of uniform ellipticity. In the variational formulation, uniform ellipticity plays a major role in establishing the coercivity of the bilinear form corresponding to the differential operator. For many problems of practical interest, however, the ellipticity can degrade. This means that the ellipticity constant α' in (10.3) becomes small. We shall see in this Chapter, that this can be especially disastrous for advection-diffusion problems.

15.1 Convection-Diffusion Problems

We consider the elliptic problem

$$\begin{aligned} -\nabla \cdot (A\nabla u) + \nabla \cdot (\mathbf{b}u) &= f, \qquad && x \in \Omega \\ u &= 0, && x \in \partial\Omega \end{aligned} \tag{15.1}$$

where $f \in L^2(\Omega)$ is a given function, and $A = (a_{ij})_{i,j=1,\ldots,d}$ with $a_{ij} \in L^\infty(\Omega)$ is a uniformly elliptic operator (cf. Definition 10.1). Furthermore, let $\mathbf{b} = (b_1, \ldots, b_d)^T$, where the b_i are smooth functions. We assume that $\nabla \cdot \mathbf{b} = 0$. This will be true if the vector $\mathbf{b}$ represents the velocity vectors of an incompressible flow field. Equation (15.1) arises, for example, when modeling the concentration of a solute in a flow field, subject to both convective transport with the fluid, and diffusion by Fick's law.

Using our standard approach for elliptic problems with homogeneous boundary conditions, we pose the problem in $H_0^1(\Omega)$. This means we find the weak solution $u \in H_0^1(\Omega)$, such that

$$\underbrace{\int_\Omega (\nabla v)^T A \nabla u dx + \int_\Omega \nabla \cdot (\mathbf{b}u) v dx}_{a(u,v)} = \underbrace{\int_\Omega f v dx}_{l(v)}, \qquad \forall v \in H_0^1(\Omega). \tag{15.2}$$

(We may or may not integrate the second term by parts. Only first derivatives appear both ways.) Using the Lax-Milgram Theorem 11.4, it is easy to show that the problem is well-posed. In particular, we can show that the bilinear form on the left-hand side of (15.2) is continuous and coercive. Continuity of the right-hand side is clear, as soon as $f \in L^2(\Omega)$. To show that the bilinear form $a(\cdot,\cdot)$ is coercive we first note that if $\nabla \cdot \mathbf{b} = 0$, we can write

$$a(v,v) = \int_\Omega (\nabla v)^T A \nabla v dx + \frac{1}{2} \int_\Omega \nabla \cdot (\mathbf{b} v^2) dx.$$

Using the divergence theorem and the boundary conditions, we see that the second term vanishes. The uniform ellipticity of the differential operator implies that there exists a positive constant α' such that

$$a(v,v) \geq \alpha' \|\nabla v\|_{0,\Omega}^2 \equiv \alpha' |v|_{1,\Omega}^2. \tag{15.3}$$

Using the Poincaré-Friedrichs inequality (see also Example 11.1), we can establish coercivity, i.e.

$$a(v,v) \geq \alpha \|v\|_{1,\Omega}^2, \qquad \text{where} \quad \alpha = c\alpha' \ (c > 0), \tag{15.4}$$

Problem (15.2) possesses a unique solution, and we may define the Ritz Galerkin method by finding an approximate solution $u_h \in S_h \subset H_0^1(\Omega)$, where S_h is a finite-dimensional subspace, such that

$$a(u_h, v_h) = l(v_h), \qquad \forall v_h \in S_h.$$

Recall from the general theory that we have an optimality statement (cf. Theorem 12.2) which asserts

$$\|u - u_h\|_{1,\Omega} \leq \frac{C}{\alpha} \inf_{v_h \in S_h} \|u - v_h\|_{1,\Omega}.$$

The coercivity constant $\alpha \equiv c\alpha'$ becomes very small if α' is very small. The quality of the error estimate may deteriorate in this case, so that we may get small errors only for excessively fine meshes. An alternative viewpoint is to look at the dependence of the solution on the data. By definition of the Galerkin method and applying a few straightforward estimates, starting from coercivity, we have

$$\alpha \|u_h\|_{1,\Omega}^2 \leq a(u_h, u_h) = l(u_h) = \int_\Omega f u_h dx \leq \|f\|_{0,\Omega} \|u_h\|_{0,\Omega} \leq \|f\|_{0,\Omega} \|u_h\|_{1,\Omega}$$

and hence

$$\|u_h\|_{1,\Omega} \leq \frac{1}{\alpha} \|f\|_{0,\Omega}$$

We see that the solution does depend continuously on the data, but the constant of continuity becomes very large for $\alpha = c\alpha' \ll 1$. (Recall that this implies that small changes in the data could lead to significant differences in the solution. If the constant of continuity is *extremely* small, this could even be true for such changes caused by round-off errors, or errors coming from numerical approximation of the integrals.)

We now have some reason to believe that the problem may become ill-behaved if the ellipticity of the differential operator degrades. To see the implications this has for the numerical approximation by the finite element method, in particular for advection-diffusion problems, we first consider the simple one-dimensional example

$$\begin{aligned} -\varepsilon u'' + bu' &= 0, \qquad x \in (0,1), \\ u(0) &= 0, \\ u(1) &= 1, \end{aligned} \tag{15.5}$$

with constants $\varepsilon, b > 0$. The prime denotes differentiation with respect to the single independent variable x. Well-posedness can be shown in a straightforward manner, using our standard theory for inhomogeneous boundary conditions (see section 14.1). You may verify by direct differentiation that the exact solution is given by

$$u(x) = \frac{e^{\frac{bx}{\varepsilon}} - 1}{e^{\frac{b}{\varepsilon}} - 1}.$$

Define the approximation space

$$V_D \quad := \quad \{v \in H^1(0,1) : v(0) = 0, v(1) = 1\}.$$

Using again standard procedure for inhomogeneous problems, in that the approximation space for u incorporates the Dirichlet boundary conditions, while the test space uses homogeneous boundary conditions, the variational formulation reads: Find $u \in V_D$, such that

$$\varepsilon \int_0^1 u'v' dx + b \int_0^1 u'v dx = 0, \qquad \forall v \in V_0 \equiv H_0^1(0,1).$$

The Ritz-Galerkin method then solves for $u_h \in V_{h,D} \subset V_D$, such that

$$\varepsilon \int_0^1 u_h' v_h' dx + b \int_0^1 u_h' v_h dx = 0, \qquad \forall v_h \in V_{h,0} \subset V_0, \tag{15.6}$$

where

$$\begin{aligned} V_{h,D} &:= \{v \in C([0,1]) : v|_{I_i} \in P^1(I_i),\, i = 1, \dots N;\ v(0) = 0, v(1) = 1\}, & (15.7)\\ V_{h,0} &:= \{v \in C([0,1]) : v|_{I_i} \in P^1(I_i),\, i = 1, \dots N;\ v(0) = 0, v(1) = 0\}. & (15.8)\end{aligned}$$

Here $P^1(I_i)$ is the space of polynomials of degree one on the interval $I_i = [x_{i-1}, x_i]$, and $x_i = ih \in \mathcal{G}_h$ corresponds to the grid

$$\mathcal{G}_h := \{ih : i = 0, \dots, N; hN = 1\}. \tag{15.9}$$

Using the basis $\phi_k \in V_{h,0}$ with $\phi_k(x_i) = \delta_{ki}$ for $k = 1, \dots, N-1$, and $i = 0, \dots, N$, we can write $u_h \in V_{h,D}$ as

$$u_h = \sum_{k=1}^{N-1} c_k \phi_k(x) + \phi_b,$$

where the boundary function

$$\phi_b = \begin{cases} \frac{x - x_{N-1}}{x_N - x_{N-1}} & x_{N-1} \le x \le x_N \\ 0 & \text{otherwise} \end{cases}.$$

has been used to enforce the boundary condition at $x = 1$. Note that the boundary treatment does not introduce any additional degrees of freedom, as the boundary value is fixed. Only the right-hand side of the problem will be affected. We obtain the discrete system

$$\sum_{k=1}^{N-1} c_k \left(\varepsilon \int_0^1 \phi_k' \phi_l' dx + b \int_0^1 \phi_k' \phi_l dx \right) + \varepsilon \int_0^1 \phi_b' \phi_l' dx + b \int_0^1 \phi_b' \phi_l dx = 0, \qquad l = 1, \dots, N-1.$$

Working out the elementary integrals one obtains

$$A\mathbf{u} = \mathbf{f},$$

where $\mathbf{u} = (c_1, \dots, c_{N-1})^T$. Note that by the nature of the chosen basis, we have $c_k = u_h(x_k)$, i.e. the kth coefficient is the value of the numerical solution at node x_k. We can write the linear system as

$$\left\{ \frac{\varepsilon}{h} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & & \ddots & & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} + \frac{b}{2} \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & 1 & & \\ & & \ddots & & \\ & & -1 & 0 & 1 \\ & & & -1 & 0 \end{pmatrix} \right\} \mathbf{u} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \frac{\varepsilon}{h} - \frac{b}{2} \end{pmatrix}.$$

Note that this is the same equation as that obtained by a finite difference discretization using a central approximation both for the second derivative and the first derivative. We define the *Péclet number* as

$$Pe = \frac{bh}{2\varepsilon},$$

and rewrite the system [1]

$$\left\{ \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & & \ddots & & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} + Pe \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & 1 & & \\ & & \ddots & & \\ & & -1 & 0 & 1 \\ & & & -1 & 0 \end{pmatrix} \right\} \mathbf{u} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 - Pe \end{pmatrix}. \tag{15.10}$$

[1] It is curious to note that for $Pe = 1$ the right-hand side of the system vanishes, and the trivial solution is obtained. Note, however, that we solve only for the interior degrees of freedom. The entire solution will not be zero in the end, due to the presence of the boundary function.

In the case of vanishing convection, $\mathbf{b} = 0$, problem (15.5) states $u_{xx} = 0$. The solution is a straight line, connecting the boundary values at the end points. The Ritz-Galerkin method is exact for this case. On the other hand, for vanishing diffusion, $\varepsilon = 0$, eq. (15.5) reads $u_x = 0$. Note that the PDE is not well posed as a boundary value problem for this case. Setting $u_x = 0$ requires u to be constant, which is in conflict with the boundary conditions. We say the differential equation (15.5) is *singularly* perturbed in ε, because in the limit $\varepsilon = 0$ the equation changes type, and needs different boundary conditions. For very small values of ε the solution of (15.5) is nearly constant for almost the entire length of the interval $(0, 1)$ before exhibiting a strong boundary layer near $x = 1$ to recover the boundary condition $u(1) = 1$. For N odd and $\varepsilon = 0$, one can find the solution of the linear system (15.10) by inspection: We have $u_h(x_i) = 0$ for all even grid indices, and $u_h(x_i) = 1$ for all odd indices. Even though the PDE problem is not well posed as stated, this is the numerical solution that is approached for very small values of ε, for which the PDE is formally well-posed, but very ill-conditioned (See fig. 15.1). It is interesting to note that we can write the analytical solution to the system (15.10) in terms of the Pectlet number, as long as $Pe \neq 1$,

$$c_j \equiv u_h(x_j) = \frac{\left(\frac{1+Pe}{1-Pe}\right)^j - 1}{\left(\frac{1+Pe}{1-Pe}\right)^N - 1}, \qquad j = 1, \ldots, N-1.$$

It should be observed that the solution is oscillatory for $Pe > 1$, and for $Pe \to \infty$ approaches the fully decoupled case found for the hyperbolic case. Fig. 15.1 shows an illustration for a *fixed* grid, and different ratios of the Peclét number.

Clearly, the Ritz-Galerkin approximation has problems in the convection-dominated case $Pe \gg 1$. A way of overcoming these problems is to increase the number of points in the grid to force $Pe \approx 1$. However, for many problems it is not always possible to increase the number of grid points at will, because this also leads to significantly increased computational cost. In the next sections we investigate other ways to get good solutions in the convection-dominated regime by introducing *stabilization* terms.

15.2 Artificial Diffusion

The Ritz-Galerkin scheme (15.6) requires a Péclet number of order unity for adequate resolution. In principle, this can be enforced by mesh refinement. For convection dominated problems, however, this may lead to excessive computational cost. In this section we discuss another way to stabilize the finite element approximation, i.e. by increasing the viscosity rather than the mesh size. To the approximation (15.6), we add an additional diffusion term, to obtain

$$(\varepsilon + \varepsilon_h) \int_0^1 u_h' v_h' dx + b \int_0^1 u_h' v_h \, dx = 0,$$

where the ε_h is termed the *artificial diffusion coefficient*. This is tantamount to producing a Galerkin approximation to a modified equation that contains more diffusion. Of course, simply adding more diffusion has implications for the accuracy of the approximation. Firstly, we note the obvious constraint that the extra viscosity has to vanish in the limit $h \to 0$ (otherwise the discretization is inconsistent), and show that viscosity of order h is sufficient to enforce a Péclet number of order unity. Choose

$$\varepsilon_h = \frac{\gamma}{2} hb, \tag{15.11}$$

where γ is a positive constant to be specified. We can define an *effective Peclet* number as

$$Pe^{(eff)} = \frac{bh}{2(\varepsilon + \varepsilon_h)} \leq \frac{bh}{2\varepsilon_h} = \frac{1}{\gamma}. \tag{15.12}$$

If $\gamma \geq 1$, we have a sufficiently small Péclet number. The downside is that this approach cannot be better than first order accurate. (Keep in mind we add an artificial diffusion term of order

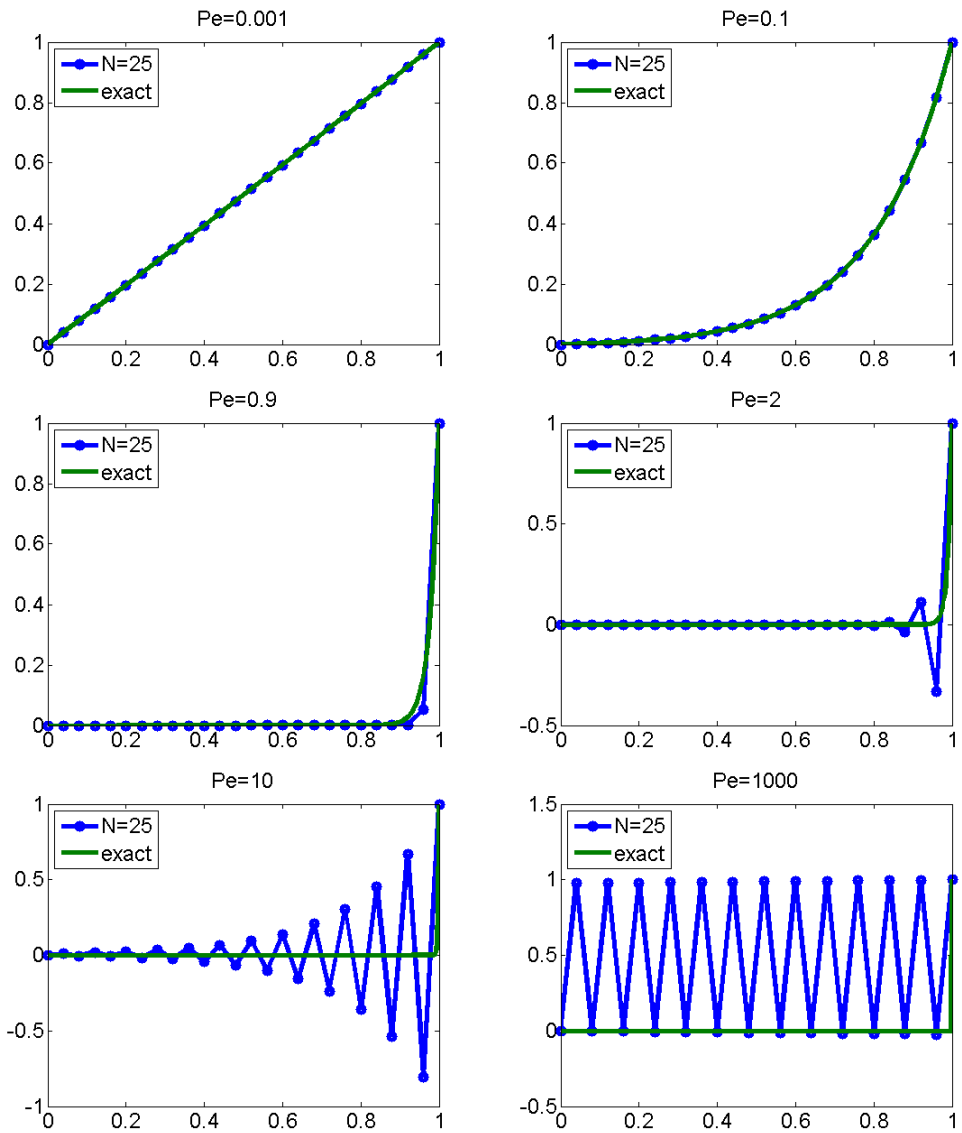

Figure 15.1: Solution of eq. (15.10) with fixed grid (N=25), corresponding to varying ratio of ε/b. For $Pe \to 0$ the solution converges to the purely elliptic solution (straight line), while for $Pe \to \infty$ the numerical solution approaches the decoupled solution $u_h(x_i) = 0$ for i even, and $u_h(x_i) = 1$ for i odd. This solution bears no resemblance to the exact solution.

h, which shouldn't be there!) This is actually consistent with the upwind concept that we used in the finite difference context, which also could be characterized as introducing an artificial viscosity of the order of the mesh spacing. Refer to the discussion on the Upwind scheme for the advection equation in Section 6.2.

What is the best choice regarding the parameter γ? If we follow (15.12), depending on the size of the *physical* diffusion ε, we may have overestimated the scaling of ε_h, necessary to obtain Péclet number of unity. Of course by simply setting $\varepsilon_h = 0$ for $Pe \leq 1$, and being a bit more careful with inequality (15.12), it is a simple matter to determine γ such that

$$Pe^{(eff)} = \begin{cases} 1 & Pe > 1, \\ Pe & Pe \leq 1. \end{cases} \tag{15.13}$$

This is shown as the "obvious" choice in Fig. 15.2.

An interesting fact emerges for the special choice $\gamma = \coth(Pe) - 1/Pe$. It can be shown that this choice gives the *exact* solution of the problem at each node. The effective Peclet number for this choice is given by $Pe^{eff} = \tanh(Pe)$. (A practical note: This means that we may just replace Pe with $\tanh(Pe)$ in (15.10), to get the exact solution at each node.) This is shown as "perfect" stabilization in Fig. 15.2.

Of course, being able to obtain the exact solution by adjusting the stabilization parameter is an artifact of this simple $1D$ model problem. However, one may use (and people have used) the simple $1D$ case as a guideline for determining in a reasonable manner the diffusion parameters in more general settings.

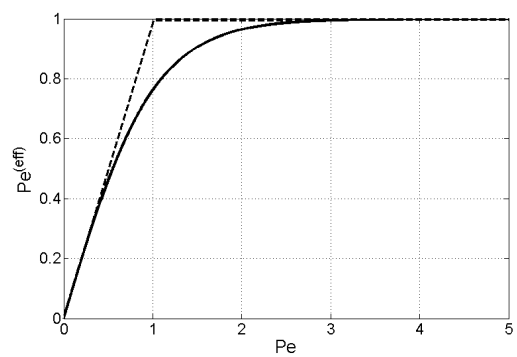

Figure 15.2: The effective Peclet number as a function of the actual Peclet number. (--): The "obvious" choice (15.13). (—): The "perfect" stabilization $\gamma = \coth(Pe) - 1/Pe$.

We now extend the artificial diffusion approach to the multidimensional case by adding additional diffusion to the standard Ritz-Galerkin approximation of eq. (15.1). We introduce a diffusion tensor A^δ, and replace eq. (15.2) with

$$\underbrace{\int_\Omega (\nabla v_h)^T A \nabla u_h dx + \int_\Omega \nabla \cdot (\mathbf{b} u_h) v_h dx}_{a(u_h, v_h)} + \underbrace{\int_\Omega (\nabla v_h)^T A^\delta \nabla u_h dx}_{a^\delta(u_h, v_h)} = \int_\Omega f v_h dx. \tag{15.14}$$

Here $a(\cdot,\cdot)$ is the same bilinear form as in (15.2). Of course, the simplest stabilization scheme is again to choose scalar diffusion, exactly as in the one-dimensional case. Assume we have based a finite-dimensional space S_h on a given mesh $\mathcal{T}_h := \{T\}$. (For example, consider the piecewise linear approximation discussed in chapter 13.) Then we can set for each mesh element T

$$A^\delta = \varepsilon_h I, \qquad \varepsilon_h|_T = \gamma \frac{h_T |\mathbf{b}|}{2}, \tag{15.15}$$

where h_T is the diameter of the mesh element T. We may use the same scaling for γ as in the one-dimensional case. (Of course for the multidimensional problem we can no longer expect to get the exact solution.) We have used I to denote the identity tensor.

Using Eq. (15.15) leads to a very diffusive scheme, and much better results can be achieved by applying diffusion only in direction of the streamlines. This means, whenever $\mathbf{b} \cdot \nabla u_h = 0$, no diffusion should be present. It makes sense to scale the diffusion with this term: We have seen that the presence of a dominant convection term can cause wild oscillations if the diffusion becomes small. But if $\mathbf{b} \cdot \nabla u_h = 0$, the convective part of (15.2) is zero, and it stands to reason that in this case no extra diffusion should be necessary.[2] To achieve the desired scaling, we replace the identity in eq. (15.15) with the outer product $\hat{\mathbf{b}}\hat{\mathbf{b}}^T$, i.e.

$$\hat{\mathbf{b}}\hat{\mathbf{b}}^T = \begin{pmatrix} \hat{b}_1\hat{b}_1 & \hat{b}_1\hat{b}_2 \\ \hat{b}_2\hat{b}_1 & \hat{b}_2\hat{b}_2 \end{pmatrix}, \qquad \hat{\mathbf{b}} = \frac{\mathbf{b}}{|\mathbf{b}|},$$

and set

$$A^\delta = \varepsilon_h \hat{\mathbf{b}}\hat{\mathbf{b}}^T. \tag{15.16}$$

By direct computation we easily verify

$$(\hat{\mathbf{b}}\hat{\mathbf{b}}^T)\nabla u_h = \frac{1}{|\mathbf{b}|^2} \begin{pmatrix} b_1 b_1 & b_1 b_2 \\ b_2 b_1 & b_2 b_2 \end{pmatrix} \begin{pmatrix} (u_h)_{x_1} \\ (u_h)_{x_2} \end{pmatrix} = \frac{1}{|\mathbf{b}|^2} \begin{pmatrix} b_1(\mathbf{b} \cdot \nabla u_h) \\ b_2(\mathbf{b} \cdot \nabla u_h) \end{pmatrix}.$$

Thus the scalar term in (15.16) will give the appropriate dependence on the parameters ε, h, and the size of the convection velocity $|\mathbf{b}|$, while the tensor $\hat{b}_i\hat{b}_j$ ensures that viscosity vanishes whenever $\mathbf{b} \cdot \nabla u_h = 0$.

Adapting the artificial viscosity to act only in direction of the streamlines is certainly an improvement compared to the simple scalar approach. However, what is lost by the explicit modification of the equations, is the variational consistency (or strong consistency) which we established first for the Galerkin approximation. Indeed, the exact solution to the advection-diffusion problem (15.1) will not satisfy the variational formulation (15.14). Substituting the exact solution u in (15.14), we obtain

$$a_h(u, v_h) - l(v_h) = a^\delta(u, v_h) \neq 0.$$

While asymptotic consistency clearly holds, as $a^\delta(u_h, v_h) \to 0$ for $h \to 0$, Galerkin orthogonality is lost. The key to overcoming this deficiency is to first realize that the artificial diffusion can equivalently be introduced by modifying the test function. Indeed, if we write

$$\int_\Omega (\nabla v_h)^T A \nabla u_h dx + \int_\Omega \mathbf{b} \cdot \nabla u_h \widetilde{v_h} dx = \int_\Omega f v_h dx,$$

where

$$\widetilde{v_h}|_T = v_h + \frac{\varepsilon_h|_T}{|\mathbf{b}|} \hat{\mathbf{b}} \cdot \nabla v_h,$$

we again obtain the same bilinear form as in (15.14). (We have used $\nabla \cdot \mathbf{b} = 0$.) Restoring strong consistency is now established by modifying *all* test functions in this manner.

15.3 Strongly Consistent Methods

In this section we derive methods that do not destroy the variational consistency of the Ritz-Galerkin formulation. First, for simplicity, we take $A = \varepsilon I$ in (15.1), where $\varepsilon > 0$ is some (small) constant. Out model problem then becomes

$$\underbrace{-\varepsilon \nabla^2 u + \nabla \cdot (\mathbf{b}u)}_{L(u)} = f.$$

[2] Recall that we have assumed $\nabla \cdot \mathbf{b} = 0$, so $\nabla \cdot (\mathbf{b}u) = \mathbf{b} \cdot \nabla u$. Even if $\nabla \cdot \mathbf{b} \neq 0$, by product rule we can write $\nabla \cdot (\mathbf{b}u) = \mathbf{b} \cdot \nabla u + (\nabla \cdot \mathbf{b})u$, where the second term is a reaction term (i.e. proportional to u), and not a convection term proportional to the first derivative of u.

Again we assume $\nabla \cdot \mathbf{b} = 0$. Then the differential operator can be written equivalently

$$L(u) = -\varepsilon \nabla^2 u + \mathbf{b} \cdot \nabla u.$$

We split $L(u)$ as

$$\hat{L} = \mathbf{b} \cdot \nabla u,$$
$$\overline{L} = -\varepsilon \nabla^2 u.$$

Recall the weak form of the equation $L(u) = f$, given in eq. (15.2). The key to the stabilized methods considered in this section is to use modified test functions,

$$v + \delta(\hat{L}(v) + \rho \overline{L}(v)), \tag{15.17}$$

where δ is a mesh-dependent parameter, and ρ is a constant. We take $\delta = const.$ such that [3]

$$\delta = c' \frac{h}{|\mathbf{b}|}, \qquad h = \max_{T \in \mathcal{T}_h} h_T. \tag{15.18}$$

The normalization is for later convenience. We consider schemes with

- $\rho = 0$: Streamline-Diffusion method, also known as SUPG (Streamline-Upwind / Petrov-Galerkin).
- $\rho = 1$: Galerkin / Least-squares method (GALS)

Below we prove a stability result for the case $\rho = 0$. The case $\rho = 1$ proceeds similarly. We generally assume that we are dealing with the underresolved case

$$Pe = \frac{|\mathbf{b}|h}{2\varepsilon} > 1,$$

We consider the streamline diffusion scheme, i.e. $\rho = 0$ in (15.17). Assume for the moment that v is a smooth function with compact support and write

$$\int_\Omega L(u)(v + \delta \hat{L}(v))dx = \int_\Omega f(v + \delta \hat{L}(v))dx.$$

Integrate by parts to obtain

$$\underbrace{\varepsilon \int_\Omega (\nabla v)^T \nabla u dx + \int_\Omega \mathbf{b} \cdot \nabla u \, v dx}_{a(u,v)} + \int_\Omega \delta \hat{L}(v)(L(u) - f)dx = \underbrace{\int_\Omega f v dx}_{l(v)}$$

We see that the variational formulation still makes sence for $u, v \in H_0^1(\Omega)$, and furthermore, for smooth enough solutions, $u \in H^2(\Omega)$, the weak form is seen to be the same as (15.2) (as u satisfies $L(u) - f = 0$). Consider now $u_h, v_h \in S_h \subset H_0^1(\Omega)$. Instead of using the Ritz-Galerkin approximation, we build the discretization on the augmented form we have just derived:

$$\underbrace{\varepsilon \int_\Omega (\nabla v_h)^T \nabla u_h dx + \int_\Omega \mathbf{b} \cdot \nabla u_h \, v_h dx}_{a(u_h,v_h)} + \underbrace{\int_\Omega \delta \hat{L}(v_h) L(u_h) dx}_{a^\delta(u_h,v_h)} = \underbrace{\int_\Omega f v_h dx}_{l(v_h)} + \underbrace{\int_\Omega f \delta \hat{L}(v_h) dx}_{l^\delta(v_h)} \tag{15.19}$$

[3] As in the previous section, the parameter can be defined locally in each mesh element T using the diameter h_T. The results derived in this section stay fundamentally the same, however, and so we opt to use a constant δ for notational convenience.

where $a(u,v)$ is the same as in the standard Galerkin approximation (15.2). In summary, we can re-write the problem

$$a_h(u_h, v_h) = l_h(v_h),$$

where

$$\begin{aligned} a_h(u_h, v_h) &= a(u_h, v_h) + a^\delta(u_h, v_h), \\ l_h(v_h) &= l(v_h) + l^\delta(v_h). \end{aligned}$$

If the exact solution satisfies $Lu = f$, then the method is strongly consistent, as $a^\delta(u, v_h) - l^\delta(v_h) = 0$ for all $v_h \in S_h$. We can write the streamline-diffusion part of the bilinear form as[4]

$$a^\delta(u_h, v_h) = -\varepsilon \sum_{T \in \mathcal{T}_h} \int_T \nabla^2 u_h \delta \hat{L}(v_h) dx + \int_\Omega \mathbf{b} \cdot \nabla u_h \delta \hat{L}(v_h) dx \tag{15.20}$$

To obtain an error estimate, we proceed in the following way. Suppose we can show, for some suitable norms $\|\cdot\|_{SUPG}$ and $\|\cdot\|_\star$, both discrete coercivity and semi-discrete boundedness, meaning the existence of positive constants α, C such that for $u_h, v_h \in S_h$, and functions $u \in H^2(\Omega)$ there holds

$$a_h(v_h, v_h) \geq \alpha \|v_h\|_{SUPG}^2 \tag{15.21}$$

$$a_h(u, v_h) \leq C \|u\|_\star \|v_h\|_{SUPG}. \tag{15.22}$$

Then we may use the triangle inequality on the error, i.e. if u is the exact solution to (15.2), and u_h is the SUPG approximation satisfying (15.19), we can write

$$\|u - u_h\|_{SUPG} \leq \|u - \mathcal{I}_h u\|_{SUPG} + \|u_h - \mathcal{I}_h u\|_{SUPG}, \tag{15.23}$$

where $\mathcal{I}_h$ is the interpolation operator into the space S_h. Using Galerkin orthogonality (implied by strong consistency), and the estimates (15.21) and (15.22), we can reduce the error estimate to a bound on the *interpolation error*. This follows from the following estimate, with which we can re-write the second term in (15.23) as

$$\begin{aligned} \|u_h - \mathcal{I}_h u\|_{SUPG}^2 &\leq \alpha^{-1} a_h(u_h - \mathcal{I}_h u, u_h - \mathcal{I}_h u) \\ &= \underbrace{a_h(u_h - u, u_h - \mathcal{I}_h u)}_{=0} + a_h(u - \mathcal{I}_h u, u_h - \mathcal{I}_h u) \\ &= a_h(u - \mathcal{I}_h u, u_h - \mathcal{I}_h u) \\ &\leq \alpha^{-1} C \|u - \mathcal{I}_h u\|_\star \|u_h - \mathcal{I}_h u\|_{SUPG}, \end{aligned}$$

This implies

$$\|u_h - \mathcal{I}_h u\|_{SUPG} \leq \alpha^{-1} C \|u - \mathcal{I}_h u\|_\star. \tag{15.24}$$

Now both terms on the right-hand side of (15.23) can be bound by the interpolation error estimate of section 13.

All we need to do, is to show the validity of (15.21) and (15.22). Indeed, we can prove the statements, as the next two theorems show. First we note (without proof) the following estimate, which holds for all $v_h \in P^k(T)$ (polynomials of degree k):

$$h_T^2 \int_T (\nabla^2 v_h)^2 dx \leq c_{0,T} \int_T |\nabla v_h|^2 dx = c_{0,T} \|\nabla v_h\|_{0,T}^2 = c_{0,T} |\nabla v_h|_{1,T}, \tag{15.25}$$

which we use in the proof of the following theorem. We assume without loss of generality that $Pe_T > 1$ for all elements. Keep in mind, that for the resolved case $Pe_T \leq 1$ one does not need any stabilization. In order to avoid case distinctions, we assume the underresolved case globally.

[4]Ssince we pose the discrete problem in $S_h \subset H_0^1(\Omega)$, we prefer to write integrals involving second derivatives as summation over element-integrals. For example, for our standard piecewise linear approximation space on triangles, $S_h \not\subset H^2(\Omega)$, as we only have continuity across triangles, but *not* continuous derivatives. However, on each triangle the solution is in any Sobolev space (polynomials!). Hence we write $\sum_T \int_T \nabla^2 u_h dx$ instead of $\int_\Omega \nabla^2 u dx$.

Theorem 15.1 *Assume $Pe_T > 1$, $\forall T \in \mathcal{T}_h$, and let the constant c' in* (15.18) *satisfy*

$$c' \leq c_0^{-1}, \qquad c_0 := \max_T c_{0,T} \tag{15.26}$$

where $c_{0,T}$ is the constant defined in inequality (15.25)*. Then the bilinear form defined in* (15.19) *satisfies*

$$a_h(v_h, v_h) \geq \frac{1}{2} \|v_h\|_{SUPG}^2$$

where

$$\|v_h\|_{SUPG}^2 := \varepsilon \|\nabla v_h\|_{0,\Omega}^2 + \delta \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega}^2$$

Proof For the Ritz-Galerkin part of the bilinear form (15.19) we obtain immediately (cf. eq. (15.4))

$$a(v_h, v_h) = \varepsilon \|\nabla v_h\|_{0,\Omega}^2 .$$

Noting that $\mathbf{b} \cdot \nabla u_h = \hat{L}(u_h)$, we obtain from the streamline-diffusion part (15.20) of the bilinear form

$$a^\delta(v_h, v_h) = -\varepsilon \sum_{T \in \mathcal{T}_h} \int_T \delta \nabla^2 v_h \hat{L}(v_h) dx + \delta \|\mathbf{b} \cdot \nabla v_h\|^2 .$$

If the piecewise linear approximation introduced in section 13 is used, the first term is actually zero. Otherwise, using inequality (15.25) and the scaled Young's inequality

$$|ab| \leq \frac{\gamma}{2} a^2 + \frac{1}{2\gamma} b^2, \qquad (\gamma > 0),$$

with $\gamma = 2$, $a = \varepsilon \sqrt{\delta} \nabla^2 v_h$, $b = \sqrt{\delta} \hat{L}(v_h)$, yields

$$\left| \varepsilon \sum_{T \in \mathcal{T}_h} \int_T \delta \nabla^2 v_h \hat{L}(v_h) dx \right| \leq \sum_{T \in \mathcal{T}_h} \int_T c_{0,T} \frac{\varepsilon^2 \delta}{h_T^2} |\nabla v_h|^2 + \frac{\delta}{4} \hat{L}(v_h)^2 dx.$$

With the definition of the Peclet number, and noting that we assume $Pe_T > 1$ we have

$$\varepsilon^2 \delta < \varepsilon \frac{|\mathbf{b}| h_T}{2} c' \frac{h_T}{|\mathbf{b}|} = c' \frac{\varepsilon}{2} h_T^2,$$

and hence, defining $c_0 = \max_T c_{0,T}$,

$$\left| |\varepsilon \sum_{T \in \mathcal{T}_h} \int_T \delta \nabla^2 v_h \hat{L}(v_h) dx \right| \leq \frac{\varepsilon}{2} c_0 c' \|\nabla v_h\|_{0,\Omega}^2 + \frac{\delta}{4} \|\mathbf{b} \cdot \nabla v_h\|^2$$

So finally, we can write

$$a(v_h, v_h) \geq \frac{\varepsilon}{2} (2 - c_0 c') \|\nabla v_h\|_{0,\Omega}^2 + \frac{\delta}{2} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega}^2 .$$

Noting assumption (15.26), this proves the theorem ■

Next we prove the semi-discrete continuity estimate (15.22):

Theorem 15.2 *Let $u \in H^2(\Omega)$. The bilinear form defined in* (15.19) *satisfies*

$$a_h(u, v_h) \leq C \|u\|_\star \|v_h\|_{SUPG}$$

where $\|\cdot\|_{SUPG}$ is defined as in (15.27)*, while*

$$\|u_h\|_\star = \left(\varepsilon \|\nabla u\|_{0,\Omega}^2 + \delta^{-1} \|u\|_{0,\Omega}^2 + \delta^2 \varepsilon \sum_T \left\|\nabla^2 u\right\|_{0,\Omega}^2 + \delta \|\mathbf{b} \cdot \nabla u\|_{0,\Omega}^2\right)^{\frac{1}{2}} \tag{15.27}$$

Proof By definition we have, upon integration by parts (using the boundary conditions)

$$a_h(u, v_h) = \varepsilon \int_\Omega (\nabla v_h)^T \nabla u dx - \int_\Omega u \mathbf{b} \cdot \nabla v_h dx - \sum_T \int_T \varepsilon \nabla^2 u \delta \hat{L}(v_h) dx + \delta \int_\Omega \mathbf{b} \cdot \nabla u \hat{L}(v_h)$$

Substituting $\hat{L}(v_h) = \mathbf{b} \cdot \nabla v_h$ gives

$$a_h(u, v_h) = \varepsilon \int_\Omega (\nabla v_h)^T \nabla u dx - \int_\Omega u \mathbf{b} \cdot \nabla v_h dx - \sum_T \int_T \varepsilon \nabla^2 u \delta \mathbf{b} \cdot \nabla v_h dx + \delta \int_\Omega \mathbf{b} \cdot \nabla u \mathbf{b} \cdot \nabla v_h$$

We apply the Cauchy-Schwartz inequality to each of the terms and obtain

$$\begin{aligned}
a_h(u, v_h) &\leq \varepsilon \|\nabla u\|_{0,\Omega} \|\nabla v_h\|_{0,\Omega} + \|u\|_{0,\Omega} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega} \\
&\quad + \varepsilon \delta \sum_T \left\|\nabla^2 u\right\|_{0,\Omega} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega} + \delta \|\mathbf{b} \cdot \nabla u\|_{0,\Omega} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega} \\
&= \varepsilon^{\frac{1}{2}} \|\nabla u\|_{0,\Omega} \varepsilon^{\frac{1}{2}} \|\nabla v_h\|_{0,\Omega} + \delta^{-\frac{1}{2}} \|u\|_{0,\Omega} \delta^{\frac{1}{2}} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega} \\
&\quad + \varepsilon \sum_T \delta^{\frac{1}{2}} \left\|\nabla^2 u\right\|_{0,\Omega} \delta^{\frac{1}{2}} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega} + \delta^{\frac{1}{2}} \|\mathbf{b} \cdot \nabla u\|_{0,\Omega} \delta^{\frac{1}{2}} \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega}
\end{aligned}$$

Using the discrete version of the same inequality,

$$\sum_i a_i b_i \leq \left(\sum_i a_i^2\right)^{\frac{1}{2}} \left(\sum_i b_i^2\right)^{\frac{1}{2}}$$

yields

$$\begin{aligned}
a_h(u, v_h) \leq &\left(\varepsilon \|\nabla u\|_{0,\Omega}^2 + \delta^{-1} \|u\|_{0,\Omega}^2 + \delta \varepsilon^2 \sum_T \left\|\nabla^2 u\right\|_{0,\Omega}^2 + \delta \|\mathbf{b} \cdot \nabla u\|_{0,\Omega}^2\right)^{\frac{1}{2}} \\
&\times \left(\varepsilon \|\nabla v_h\|_{0,\Omega}^2 + 3\delta \|\mathbf{b} \cdot \nabla v_h\|_{0,\Omega}^2\right)^{\frac{1}{2}} \leq C \|u\|_\star \|v_h\|_{SUPG}
\end{aligned}$$

■

As outlined above, we may now proceed from the triangle inequality (15.23) to obtain the error estimate for the SUPG method:

Theorem 15.3 *If piecewise affine linear approximation is employed, the error of the SUPG scheme satisfies*

$$\|u - u_h\|_{SUPG} = Ch^{\frac{3}{2}} |u|_{2,\Omega}$$

PROOF Recall triangle inequality (15.23):

$$\|u - u_h\|_{SUPG} \leq \|u - \mathcal{I}_h u\|_{SUPG} + \|u_h - \mathcal{I}_h u\|_{SUPG}$$

Using standard interpolation results (cf. Theorem 13.4)

$$\begin{aligned}\|u - \mathcal{I}_h u\|_{SUPG} &= \left(\varepsilon \|\nabla(u - \mathcal{I}_h u)\|_{0,\Omega}^2 + \delta \|\mathbf{b} \cdot \nabla(u - \mathcal{I}_h u)\|_{0,\Omega}^2\right)^{\frac{1}{2}} \\ &\leq C \left(\varepsilon \underbrace{|u - \mathcal{I}_h u|_{1,\Omega}^2}_{\mathcal{O}(h^2)} + \delta \underbrace{|u - \mathcal{I}_h u|_{1,\Omega}^2}_{=\mathcal{O}(h^2)}\right)^{\frac{1}{2}} = \mathcal{O}(h^{\frac{3}{2}})\end{aligned}$$

(Note that $\delta = \mathcal{O}(h)$, and since we have assumed $Pe > 1$, we can assume $\varepsilon = \mathcal{O}(h)$.) Using Theorems 15.1 and 15.2, we can estimate the second term using eq. (15.24) with $\alpha = 1/2$,

$$\begin{aligned}\frac{1}{2} \|u_h - \mathcal{I}_h u\|_{SUPG} \leq C \Big(&\varepsilon \|\nabla(u - \mathcal{I}_h u)\|_{0,\Omega}^2 + \delta^{-1} \|u - \mathcal{I}_h u\|_{0,\Omega}^2 \\ &+ \delta^2 \varepsilon \sum_T \left\|\nabla^2(u - \mathcal{I}_h u)\right\|_{0,\Omega}^2 + \delta \|\mathbf{b} \cdot \nabla(u - \mathcal{I}_h u)\|_{0,\Omega}^2\Big)^{\frac{1}{2}}\end{aligned}$$

Noting again $\delta = \mathcal{O}(h)$, and $\varepsilon = \mathcal{O}(h)$, standard interpolation estimates reveal all the terms inside the bracket to be of order h^3. After taking the square root, the result follows. ■

15.4 Further Reading

The exposition of stabilized schemes follows Quarteroni [9], Chapter 8. There, in addition to the results on SUPG schemes covered here, you'll also find results on GALS schemes.